

Practical 2

Aim: Hadoop Installation, Configuration and HDFS Operations on Windows/Linux

Problem definition:

You have recently joined DataAnalytics Inc. as a Hadoop Administrator. The organization processes large-scale customer, sales, and marketing data using the Hadoop ecosystem. Due to increasing data volumes, the company has decided to migrate its data storage infrastructure from traditional file systems to Hadoop Distributed File System (HDFS). As a Hadoop Administrator, your responsibility is to install and configure Hadoop in a Single-Node Cluster environment on Windows/Linux, verify the Hadoop services, manage HDFS storage, and perform data management operations. Management wants a fully functional Hadoop environment that can store, retrieve, and process big data efficiently. Your task is to install Hadoop, configure the Hadoop ecosystem, start Hadoop services, and perform HDFS operations using Hadoop commands.

Task:

- Hadoop Installation & Configuration

- Access Hadoop Web UI.

1. Create directories in HDFS.
2. Upload files to HDFS.
3. List HDFS files and directories.
4. Display file contents.
5. Copy files within HDFS.
6. Move files within HDFS.
7. Rename files.
8. Download files from HDFS.
9. Delete files and directories.
10. Check HDFS report.
11. Verify block information.
12. Check Hadoop cluster health.
13. Monitor Hadoop processes.
14. Check storage utilization.

Objectives:

- To understand the architecture and working of the Hadoop Distributed File System (HDFS).
- To install and configure Hadoop in a Single-Node Cluster environment.
- To start, stop, and verify Hadoop services using administrative commands.
- To access and monitor the Hadoop Web UI for cluster management.
- To perform HDFS file and directory management operations such as creating, uploading, listing, copying, moving, renaming, downloading, and deleting files.
- To monitor Hadoop cluster health, storage utilization, and running processes.
- To verify HDFS block information and generate cluster reports using Hadoop administrative commands.
- To gain practical experience in managing and maintaining a Hadoop environment for big data storage and processing.

Observation / Output

```
(tushar@Tushar-laptop)-[~]  
$ java -version  
openjdk version "21.0.11" 2026-04-21  
OpenJDK Runtime Environment (build 21.0.11+10-1-Debian)  
OpenJDK 64-Bit Server VM (build 21.0.11+10-1-Debian, mixed mode, sharing)
```

```
(tushar@Tushar-laptop)-[~]  
$ hadoop version  
Hadoop 3.4.3  
Source code repository https://github.com/apache/hadoop.git -r 9d50c6884666e794e45102260a4017bb31802e1b  
Compiled by stevel on 2026-02-13T14:23Z  
Compiled on platform linux-x86_64  
Compiled with protoc 3.23.4  
From source with checksum 2331238c4c2929e66645316a32a8613  
This command was run using /home/tushar/hadoop/share/hadoop/common/hadoop-common-3.4.3.jar
```

```
(tushar@Tushar-Laptop)-[~]
$ echo $JAVA_HOME
echo $HADOOP_HOME
echo $PATH
/usr/lib/jvm/java-21-openjdk-amd64
/home/tushar/hadoop
/home/tushar/.local/bin:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/s
we:/mnt/c/Program Files/Common Files/Oracle/Java/javapath:/mnt/c/WINDOWS/sys
/mnt/c/Program Files (x86)/Microsoft SQL Server/170/Tools/Binn:/mnt/c/Progr
r/170/DTS/Binn:/mnt/c/Users/tusha/AppData/Local/nvm:/mnt/c/nvm4w/nodejs:/mr
ograms/Python/Launcher:/mnt/c/Users/tusha/AppData/Local/Microsoft/WindowsAp
ing/npm:/mnt/c/Users/tusha/AppData/Local/Programs/cursor/resources/app/bin:/
nt/c/Users/tusha/AppData/Local/Kingsoft/WPS Office/12.1.0.27458/clitool:/ho
ar/hadoop/bin:/home/tushar/hadoop/sbin
```

```
(tushar@Tushar-laptop)-[~]
$ cat ~/hadoop/etc/hadoop/core-site.xml
<?xml version="1.0" encoding="UTF-8"?>
<?xml-stylesheet type="text/xsl" href="configuration.xsl"?>
<!--
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distributed under the License is distributed on an "AS IS" BASIS,
WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
See the License for the specific language governing permissions and
limitations under the License. See accompanying LICENSE file.
-->
<!-- Put site-specific property overrides in this file. -->

<configuration>
<property>
  <name>fs.defaultFS</name>
  <value>hdfs://localhost:9000</value>
</property>
</configuration>
```

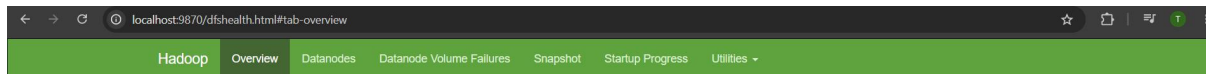
```
(tushar@Tushar-laptop)-[~]
$ cat ~/hadoop/etc/hadoop/hdfs-site.xml
<?xml version="1.0" encoding="UTF-8"?>
<?xml-stylesheet type="text/xsl" href="configuration.xsl"?>
<!--
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distributed under the License is distributed on an "AS IS" BASIS,
WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
See the License for the specific language governing permissions and
limitations under the License. See accompanying LICENSE file.
-->
<!-- Put site-specific property overrides in this file. -->

<configuration>
<property>
  <name>dfs.replication</name>
  <value>1</value>
</property>
<property>
  <name>dfs.name.dir</name>
  <value>file:///home/tushar/hadoopdata/hdfs/namenode</value>
</property>
<property>
  <name>dfs.data.dir</name>
  <value>file:///home/tushar/hadoopdata/hdfs/datanode</value>
</property>
</configuration>
```

```
(tushar@Tushar-laptop)-[~]
$ jps
4290 NameNode
20978 Jps
4694 SecondaryNameNode
7786 ResourceManager
4429 DataNode
7918 NodeManager
```



Overview 'localhost:9000' (✓active)

Started:	Thu Aug 06 22:07:09 +0530 2026
Version:	3.4.3, r9d50c6884666e794e45102260a017bb31802e1b
Compiled:	Fri Feb 13 19:53:00 +0530 2026 by steverl from branch-3.4.3
Cluster ID:	CID-5f6ea5c8-00d3-49ad-913c-88b27f6e47e7
Block Pool ID:	BP-1266567477-127.0.1.1-1784169044755

Summary

Security is off.

Safemode is off.

7 files and directories, 2 blocks (2 replicated blocks, 0 erasure coded block groups) = 9 total filesystem object(s).

Heap Memory used 68.93 MB of 105 MB Heap Memory. Max Heap Memory is 1.9 GB.

Non Heap Memory used 59.53 MB of 61.63 MB Committed Non Heap Memory. Max Non Heap Memory is <unbounded>.

Cluster Metrics

Apps Submitted	0	Apps Pending	0	Apps Running	0	Apps Completed	0	Containers Running	0	Used Resources	<memory 0 B, vCores 0>	Total Resources	<memory 8 GB, vCores 8>	Reserved Resources	<memory 0 B, vCores 0>
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Cluster Nodes Metrics

Active Nodes	0	Decommissioning Nodes	0	Decommissioned Nodes	0	Lost Nodes	0	Unhealthy Nodes	0
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Scheduler Metrics

Scheduler Type	Capacity Scheduler	Scheduling Resource Type	[memory-mb (unit-M), vcores]	Minimum Allocation	<memory 1024, vCores 1>	Maximum Allocation	<memory 8192, vCores 4>	Maximum Cluster Application Priority	0	Scheduler Busy %	0	RM Dispatcher EventQ	0
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Showing 0 to 0 of 0 entries

```
(tushar@Tushar-laptop)-[~/MarketingAnalytics]
$ start-dfs.sh
Starting namenodes on [localhost]
Starting datanodes
Starting secondary namenodes [Tushar-laptop]
```

```
(tushar@Tushar-laptop)-[~/MarketingAnalytics]
$ start-yarn.sh
Starting resourcemanager
Starting nodemanagers
```

```
(tushar@Tushar-laptop)-[~/MarketingAnalytics]
$ jps
4321 NameNode
5009 ResourceManager
4481 DataNode
5124 NodeManager
5541 Jps
4728 SecondaryNameNode
```

```
(tushar@Tushar-laptop)-[~/MarketingAnalytics]
$ hdfs dfs -ls /
Found 1 items
drwxr-xr-x - tushar supergroup          0 2026-07-16 08:05 /input

(tushar@Tushar-laptop)-[~/MarketingAnalytics]
$ hdfs dfs -mkdir /MarketingAnalytics

(tushar@Tushar-laptop)-[~/MarketingAnalytics]
$ hdfs dfs -ls /
Found 2 items
drwxr-xr-x - tushar supergroup          0 2026-07-17 15:43 /MarketingAnalytics
drwxr-xr-x - tushar supergroup          0 2026-07-16 08:05 /input

(tushar@Tushar-laptop)-[~/MarketingAnalytics]
$ hdfs dfs -mkdir /MarketingAnalytics/RawData
hdfs dfs -mkdir /MarketingAnalytics/ProcessedData
hdfs dfs -mkdir /MarketingAnalytics/Reports

(tushar@Tushar-laptop)-[~/MarketingAnalytics]
$ hdfs dfs -ls /MarketingAnalytics
Found 3 items
drwxr-xr-x - tushar supergroup          0 2026-07-17 15:44 /MarketingAnalytics/ProcessedData
drwxr-xr-x - tushar supergroup          0 2026-07-17 15:44 /MarketingAnalytics/RawData
drwxr-xr-x - tushar supergroup          0 2026-07-17 15:44 /MarketingAnalytics/Reports
```

```
(tushar@Tushar-laptop)-[~/MarketingAnalytics]
$ echo "Sales Report for July" > sales.txt
echo "Customer Data 2026" > customers.txt

(tushar@Tushar-laptop)-[~/MarketingAnalytics]
$ hdfs dfs -put sales.txt /MarketingAnalytics/RawData
hdfs dfs -put customers.txt /MarketingAnalytics/RawData
```

```
(tushar@Tushar-laptop)-[~/MarketingAnalytics]
$ hdfs dfs -ls /MarketingAnalytics/RawData
Found 2 items
-rw-r--r-- 1 tushar supergroup          19 2026-07-17 15:49 /MarketingAnalytics/RawData/customers.txt
-rw-r--r-- 1 tushar supergroup          22 2026-07-17 15:49 /MarketingAnalytics/RawData/sales.txt
```

```
(tushar@Tushar-laptop)-[~/MarketingAnalytics]
$ hdfs dfs -cat /MarketingAnalytics/RawData/sales.txt
Sales Report for July

(tushar@Tushar-laptop)-[~/MarketingAnalytics]
$ hdfs dfs -cat /MarketingAnalytics/RawData/customers.txt
Customer Data 2026
```

```
(tushar@Tushar-laptop)-[~/MarketingAnalytics]
$ hdfs dfs -cp /MarketingAnalytics/RawData/sales.txt /MarketingAnalytics/Reports

(tushar@Tushar-laptop)-[~/MarketingAnalytics]
$ hdfs dfs -ls /MarketingAnalytics/Reports
Found 1 items
-rw-r--r-- 1 tushar supergroup          22 2026-07-17 15:58 /MarketingAnalytics/Reports/sales.txt
```

```
(tushar@Tushar-laptop)-[~/MarketingAnalytics]
$ hdfs dfs -mv /MarketingAnalytics/Reports/sales.txt /MarketingAnalytics/Reports/FinalReport.txt

(tushar@Tushar-laptop)-[~/MarketingAnalytics]
$ hdfs dfs -ls /MarketingAnalytics/Reports
Found 1 items
-rw-r--r-- 1 tushar supergroup          22 2026-07-17 15:58 /MarketingAnalytics/Reports/FinalReport.txt
```

```
(tushar@Tushar-laptop)-[~/MarketingAnalytics]
$ mkdir DownloadedFiles

(tushar@Tushar-laptop)-[~/MarketingAnalytics]
$ hdfs dfs -get /MarketingAnalytics/Reports/FinalReport.txt DownloadedFiles

(tushar@Tushar-laptop)-[~/MarketingAnalytics]
$ ls DownloadedFiles
FinalReport.txt
```

```
(tushar@Tushar-laptop)-[~/MarketingAnalytics]
$ hdfs dfs -rm /MarketingAnalytics/ProcessedData/customers.txt
Deleted /MarketingAnalytics/ProcessedData/customers.txt

(tushar@Tushar-laptop)-[~/MarketingAnalytics]
$ hdfs dfs -rm /MarketingAnalytics/Reports/FinalReport.txt
Deleted /MarketingAnalytics/Reports/FinalReport.txt

(tushar@Tushar-laptop)-[~/MarketingAnalytics]
$ hdfs dfs -rmdir /MarketingAnalytics/Reports

(tushar@Tushar-laptop)-[~/MarketingAnalytics]
$ hdfs dfs -ls /MarketingAnalytics
Found 2 items
drwxr-xr-x - tushar supergroup          0 2026-07-17 16:04 /MarketingAnalytics/ProcessedData
drwxr-xr-x - tushar supergroup          0 2026-07-17 15:58 /MarketingAnalytics/RawData
```

```
(tushar@Tushar-laptop)-[~/MarketingAnalytics]
$ hdfs dfsadmin -report
Configured Capacity: 1081101176832 (1006.85 GB)
Present Capacity: 1008726339521 (939.45 GB)
DFS Remaining: 1008726265856 (939.45 GB)
DFS Used: 73665 (71.94 KB)
DFS Used%: 0.00%
Replicated Blocks:
    Under replicated blocks: 0
    Blocks with corrupt replicas: 0
    Missing blocks: 0
    Missing blocks (with replication factor 1): 0
    Low redundancy blocks with highest priority to recover: 0
    Pending deletion blocks: 0
Erasure Coded Block Groups:
    Low redundancy block groups: 0
    Block groups with corrupt internal blocks: 0
    Missing block groups: 0
    Low redundancy blocks with highest priority to recover: 0
    Pending deletion blocks: 0

-----
Live datanodes (1):

Name: 127.0.0.1:9866 (localhost)
Hostname: Tushar-laptop.localdomain
Decommission Status : Normal
Configured Capacity: 1081101176832 (1006.85 GB)
DFS Used: 73665 (71.94 KB)
Non DFS Used: 17382514751 (16.19 GB)
DFS Remaining: 1008726233088 (939.45 GB)
DFS Used%: 0.00%
DFS Remaining%: 93.31%
Configured Cache Capacity: 0 (0 B)
Cache Used: 0 (0 B)
Cache Remaining: 0 (0 B)
Cache Used%: 100.00%
Cache Remaining%: 0.00%
Xceivers: 0
Last contact: Fri Jul 17 16:06:42 IST 2026
Last Block Report: Fri Jul 17 15:39:06 IST 2026
Num of Blocks: 2
```

```
(tushar@Tushar-laptop) [~/MarketingAnalytics]
$ hdfs fsck /MarketingAnalytics/RawData/sales.txt
Connecting to namenode via http://localhost:9870/fscck?ugi=tushar&path=%2FMarketingAnalytics%2FRawData%2Fsales.txt
FSCK started by tushar (auth:SIMPLE) from /127.0.0.1 for path /MarketingAnalytics/RawData/sales.txt at Fri Jul 17 16:07:30 IST 2026

Status: HEALTHY
Number of data-nodes: 1
Number of racks: 1
Total dirs: 0
Total symlinks: 0

Replicated Blocks:
Total size: 22 B
Total files: 1
Total blocks (validated): 1 (avg. block size 22 B)
Minimally replicated blocks: 1 (100.0 %)
Over-replicated blocks: 0 (0.0 %)
Under-replicated blocks: 0 (0.0 %)
Mis-replicated blocks: 0 (0.0 %)
Default replication factor: 1
Average block replication: 1.0
Missing blocks: 0
Corrupt blocks: 0
Missing replicas: 0 (0.0 %)
Blocks queued for replication: 0

Erasure Coded Block Groups:
Total size: 0 B
Total files: 0
Total block groups (validated): 0
Minimally erasure-coded block groups: 0
Over-erasure-coded block groups: 0
Under-erasure-coded block groups: 0
Unsatisfactory placement block groups: 0
Average block group size: 0.0
Missing block groups: 0
Corrupt block groups: 0
Missing internal blocks: 0
Blocks queued for replication: 0
FSCK ended at Fri Jul 17 16:07:30 IST 2026 in 3 milliseconds

The filesystem under path '/MarketingAnalytics/RawData/sales.txt' is HEALTHY
```

```
(tushar@Tushar-laptop)-[~/MarketingAnalytics]
$ jps
hdfs dfsadmin -report
4321 NameNode
5009 ResourceManager
4481 DataNode
5124 NodeManager
8344 Jps
4728 SecondaryNameNode
Configured Capacity: 1081101176832 (1006.85 GB)
Present Capacity: 1008726306753 (939.45 GB)
DFS Remaining: 1008726233088 (939.45 GB)
DFS Used: 73665 (71.94 KB)
DFS Used%: 0.00%
Replicated Blocks:
  Under replicated blocks: 0
  Blocks with corrupt replicas: 0
  Missing blocks: 0
  Missing blocks (with replication factor 1): 0
  Low redundancy blocks with highest priority to recover: 0
  Pending deletion blocks: 0
Erasure Coded Block Groups:
  Low redundancy block groups: 0
  Block groups with corrupt internal blocks: 0
  Missing block groups: 0
  Low redundancy blocks with highest priority to recover: 0
  Pending deletion blocks: 0

-----
Live datanodes (1):

Name: 127.0.0.1:9866 (localhost)
Hostname: Tushar-laptop.localdomain
Decommission Status : Normal
Configured Capacity: 1081101176832 (1006.85 GB)
DFS Used: 73665 (71.94 KB)
Non DFS Used: 17382514751 (16.19 GB)
DFS Remaining: 1008726233088 (939.45 GB)
DFS Used%: 0.00%
DFS Remaining%: 93.31%
Configured Cache Capacity: 0 (0 B)
Cache Used: 0 (0 B)
Cache Remaining: 0 (0 B)
Cache Used%: 100.00%
Cache Remaining%: 0.00%
Xceivers: 0
Last contact: Fri Jul 17 16:08:18 IST 2026
Last Block Report: Fri Jul 17 15:39:06 IST 2026
Num of Blocks: 2
```

```
(tushar@Tushar-laptop)-[~/MarketingAnalytics]
$ jps
8512 Jps
4321 NameNode
5009 ResourceManager
4481 DataNode
5124 NodeManager
4728 SecondaryNameNode
```

```
(tushar@Tushar-laptop)~[~/MarketingAnalytics]
$ hdfs dfs -df -h
Filesystem              Size      Used    Available    Use%
hdfs://localhost:9000  1006.9 G    56 K      939.4 G      0%

(tushar@Tushar-laptop)~[~/MarketingAnalytics]
$ df -h
Filesystem              Size      Used    Avail  Use% Mounted on
none                    3.9G      0      3.9G    0% /usr/lib/modules/6.18.33.2-microsoft-standard-WSL2
none                    3.9G    4.0K    3.9G    1% /mnt/wsl
drivers                  280G    170G    111G   61% /usr/lib/wsl/drivers
/dev/sdd                 1007G    17G    940G    2% /
none                    3.9G   156K    3.9G    1% /mnt/wslg
none                    3.9G      0      3.9G    0% /usr/lib/wsl/lib
rootfs                   3.8G    2.8M    3.8G    1% /init
none                    3.8G      0      3.8G    0% /dev
none                    3.9G   824K    3.8G    1% /run
none                    3.9G      0      3.9G    0% /run/lock
none                    3.9G      0      3.9G    0% /run/shm
none                    3.9G   936K    3.8G    1% /mnt/wslg/versions.txt
none                    3.9G   936K    3.8G    1% /mnt/wslg/doc
C:\                      280G    170G    111G   61% /mnt/c
D:\                      196G    1.4G   195G    1% /mnt/d
none                    1.0M      0      1.0M    0% /run/credentials/systemd-journald.service
none                    1.0M      0      1.0M    0% /run/credentials/getty@tty1.service
tmpfs                   779M    20K    779M    1% /run/user/1000
```

```
(tushar@Tushar-laptop)~[~/MarketingAnalytics]
$ stop-yarn.sh
Stopping nodemanagers
Stopping resourcemanager

(tushar@Tushar-laptop)~[~/MarketingAnalytics]
$ stop-dfs.sh
Stopping namenodes on [localhost]
Stopping datanodes
Stopping secondary namenodes [Tushar-laptop]

(tushar@Tushar-laptop)~[~/MarketingAnalytics]
$ jps
9318 Jps
```

Key Questions /Analysis:

1. Why is Hadoop preferred for Big Data storage?

Hadoop is preferred for Big Data storage because it can store and process very large amounts of data efficiently. It uses HDFS to distribute data across multiple machines, providing scalability and fault tolerance. Hadoop also supports parallel processing, making data analysis faster. It is cost-effective as it runs on commodity hardware. Therefore, it is widely used for Big Data applications.

2. What is the role of NameNode and DataNode in HDFS?

The NameNode is the master node that manages the metadata and directory structure of HDFS. It keeps track of file locations and block information. The DataNode stores the actual data blocks and performs read/write operations. It regularly sends heartbeat signals to the NameNode. Together, they ensure efficient and reliable data storage.

3. What is the difference between HDFS and traditional file systems?

HDFS is a distributed file system designed for storing Big Data across multiple machines, while traditional file systems store data on a single machine. HDFS provides fault tolerance through data replication and is highly scalable. Traditional file systems have limited scalability and no automatic replication. HDFS is optimized for parallel processing and large datasets.

4. What are the advantages of distributed storage?

Distributed storage improves data availability by storing data across multiple nodes. It provides fault tolerance through replication and ensures high reliability. It also allows easy scalability by adding new nodes to the cluster. Parallel access to data improves processing speed. Overall, distributed storage is efficient for handling large-scale data.

1. Why are backups important in enterprise environments?

Backups protect important business data from accidental deletion, hardware failure, ransomware attacks, and system crashes. They ensure business continuity and quick data recovery.

Supplementary Problems:

1: Configure Hadoop Pseudo-Distributed Mode

```
(tushar@Tushar-laptop)-[~]
$ start-dfs.sh
Starting namenodes on [localhost]
Starting datanodes
Starting secondary namenodes [Tushar-laptop]

(tushar@Tushar-laptop)-[~]
$ jps
4290 NameNode
4694 SecondaryNameNode
5432 Jps
4429 DataNode
```

[illegible]

2: Change HDFS replication factor and observe results

```
(tushar@Tushar-laptop)~$ hdfs dfs -setrep 2 /MarketingAnalytics/RawData/sales.txt
Replication 2 set: /MarketingAnalytics/RawData/sales.txt

(tushar@Tushar-laptop)~$ hdfs fsck /MarketingAnalytics/RawData/sales.txt
Connecting to namenode via http://localhost:9870/fsck?ugi=tushar&path=%2FMarketingAnalytics%2FRawData%2Fsales.txt
FSCK started by tushar (auth:SIMPLE) from /127.0.0.1 for path /MarketingAnalytics/RawData/sales.txt at Thu Aug 06 22:21:33 IST 2026
Isiye ye command fail hui thi.

/MarketingAnalytics/RawData/sales.txt: Under replicated BP-1266567477-127.0.1.1-1784169044755:blk_1073741826_1002. Target Replicas is 2 but found 1 live replica(s),
Status: HEALTHY
Number of data-nodes: 1
Number of racks: 1
Total dirs: 0
Total symlinks: 0

Replicated Blocks:
Total size: 22 B
Total files: 1
Total blocks (validated): 1 (avg. block size 22 B)
Minimally replicated blocks: 1 (100.0 %)
Over-replicated blocks: 0 (0.0 %)
Under-replicated blocks: 1 (100.0 %)
Mis-replicated blocks: 0 (0.0 %)
Default replication factor: 1
Average block replication: 1.0
Missing blocks: 0
Corrupt blocks: 0
Missing replicas: 1 (50.0 %)
Blocks queued for replication: 0

Erasure Coded Block Groups:
Total size: 0 B
Total files: 0
Total block groups (validated): 0
Minimally erasure-coded block groups: 0
Over-erasure-coded block groups: 0
Under-erasure-coded block groups: 0
Unsatisfactory placement block groups: 0
Average block group size: 0.0
Missing block groups: 0
Corrupt block groups: 0
Missing internal blocks: 0
Blocks queued for replication: 0
FSCK ended at Thu Aug 06 22:21:33 IST 2026 in 13 milliseconds

Data filesystem is Healthy
The filesystem under path '/MarketingAnalytics/RawData/sales.txt' is HEALTHY

(tushar@Tushar-laptop)~$ hdfs dfs -stat %r /MarketingAnalytics/RawData/sales.txt
2

(tushar@Tushar-laptop)~$
```

3: Upload a Large Dataset into HDFS

```
(tushar@Tushar-laptop)~$ hdfs dfs -mkdir /BigData

(tushar@Tushar-laptop)~$ hdfs dfs -put sales.txt /BigData
put: 'sales.txt': No such file or directory

(tushar@Tushar-laptop)~$ hdfs dfs -get /MarketingAnalytics/RawData/sales.txt

(tushar@Tushar-laptop)~$ hdfs dfs -put sales.txt /BigData

(tushar@Tushar-laptop)~$ hdfs dfs -ls /BigData
Found 1 items
-rw-r--r-- 1 tushar supergroup 22 2026-08-06 22:26 /BigData/sales.txt

(tushar@Tushar-laptop)~$
```

4: Analyze HDFS Storage Utilization

```
(tushar@Tushar-laptop)-[~]  
$ hdfs dfsadmin -report  
Configured Capacity: 1081101176832 (1006.85 GB)  
Present Capacity: 1006883250176 (937.73 GB)  
DFS Remaining: 1006883184640 (937.73 GB)  
DFS Used: 65536 (64 KB)  
DFS Used%: 0.00%  
Replicated Blocks:  
   Under replicated blocks: 1  
   Blocks with corrupt replicas: 0  
   Missing blocks: 0  
   Missing blocks (with replication factor 1): 0  
   Low redundancy blocks with highest priority to recover: 1  
   Pending deletion blocks: 0  
Erasure Coded Block Groups:  
   Low redundancy block groups: 0  
   Block groups with corrupt internal blocks: 0  
   Missing block groups: 0  
   Low redundancy blocks with highest priority to recover: 0  
   Pending deletion blocks: 0  
  
Live datanodes (1):  
Name: 127.0.0.1:9866 (localhost)  
Hostname: Tushar-laptop.localdomain  
Decommission Status : Normal  
Configured Capacity: 1081101176832 (1006.85 GB)  
DFS Used: 65536 (64 KB)  
Non DFS Used: 19225571328 (17.91 GB)  
DFS Remaining: 1006883184640 (937.73 GB)  
DFS Used%: 0.00%  
DFS Remaining%: 93.13%  
Configured Cache Capacity: 0 (0 B)  
Cache Used: 0 (0 B)  
Cache Remaining: 0 (0 B)  
Cache Used%: 100.00%  
Cache Remaining%: 0.00%  
Xceivers: 0  
Last contact: Thu Aug 06 22:28:48 IST 2026  
Last Block Report: Thu Aug 06 22:07:12 IST 2026  
Num of Blocks: 3
```

Tools /Technology used:

- Ubuntu Linux / Kali Linux (WSL)
- Hadoop 3.x
- OpenJDK 11 or above
- HDFS (Hadoop Distributed File System)
- Linux Terminal

- Bash Shell
- SSH (OpenSSH Server)
- Hadoop Web UI (NameNode – Port 9870)
- Text Editor (Nano / Vim)

Conclusion :

The Hadoop single-node cluster was successfully installed and configured on Linux. HDFS services were started, and various file management operations such as creating directories, uploading, downloading, copying, moving, renaming, and deleting files were performed successfully. The Hadoop Web UI and HDFS reports were used to verify cluster status, storage utilization, and DataNode information. This practical provided hands-on experience in Hadoop administration and demonstrated the efficient management of distributed storage using HDFS.